

SQL Data Analytics Portfolio Report

A compact summary of the SQL Server analytics project, covering the data model, analysis workflow, and portfolio skills demonstrated.

Project Scope

This project builds an analytics-ready SQL Server database for sales analysis. It starts with schema creation and data loading, then progresses through exploration, KPI analysis, segmentation, trend analysis, and reusable reporting views.

Data Model

Table	Role	Key fields
gold.dim_customers	Customer dimension	customer_key, country, gender, birthdate
gold.dim_products	Product dimension	product_key, category, subcategory, cost
gold.fact_sales	Sales fact table	order_number, order_date, sales_amount, quantity, price

Analysis Workflow

- Initialize the DataWarehouseAnalytics database and load customer, product, and sales files.
- Explore metadata, dimensions, date ranges, and core business measures.
- Analyze magnitude, ranking, time trends, cumulative growth, and contribution share.
- Segment customers by lifespan and sales value, and segment products by revenue performance.
- Create customer and product reporting views for downstream analysis.

Skills Demonstrated

- SQL Server DDL and schema creation
- Data loading with BULK INSERT
- CTEs, joins, aggregations, CASE logic, and date functions
- Window functions for ranking and period comparisons
- KPI design for sales, customers, products, recency, and monthly spend
- Portfolio documentation and report packaging

Reviewer Highlights

The strongest portfolio signal is the progression from raw tables to analysis-ready reporting views. The project demonstrates not just individual SQL queries, but a complete analytical path from setup through business-facing outputs.

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